Rank-seven Grothendieck rings: certified exclusions and models through multiplicity four

Abstract

We apply general categorification obstructions to the pinned, complete fusion-ring census of rank seven and multiplicity at most four. Every asserted exclusion has a replayable exact certificate; every asserted realization has an explicit unitary fusion-category model and an exact full-tensor identification. We include a rank-seven extension of the formal-codegree test to noncommutative rings, a global-dimension-fifteen obstruction, and complete oriented signed-sphericalization certificates for 246 rings through multiplicity four. The remaining tensors are recorded as *undecided*: the present results do not classify all rank-seven Grothendieck rings, already at multiplicity one.

1 Scope and statements

Let R have distinguished basis $(b_0 = 1, b_1, \ldots, b_6)$ and multiplication $b_i b_j = \sum_k N_{ij}^k b_k$. A *Grothendieck ring* here means the based ring of a complex fusion category; no pivotal, spherical, braided, or unitary hypothesis is placed on a hypothetical categorification. Its multiplicity is $\max_{i,j,k} N_{ij}^k$. The upstream *Fusion Ring Census* [1] provides the complete rank-seven fusion-ring list through multiplicity four. We pin its source bytes, source commit, row ordering, and their digest in the companion package. This is a categorification study of that census; the list of undecided tensors is not counted as a list of Grothendieck rings. Multiplicities at least five lie outside the pinned source and the claims of this manuscript.

Theorem 1 (Certified rank-seven status through multiplicity four)**.** *The numbers of excluded, explicitly unitarily realized, and undecided rank-seven fusion rings are given by Table 1. The negative conclusions hold for* every *complex fusion category. The sixteen positive rows have the models in Table 3, matched against their complete fusion tensors. No conclusion is drawn for the undecided rows.*

Table 1: Certified status of complete rank-seven fusion-ring census through exact multiplicity four. Undecided entries are not assumed categorifiable.

Exact multiplicity	Fusion rings	Excluded	Realized	Undecided
1	43	30	10	3
2	319	291	5	23
3	1059	1035	1	23
4	2780	2751	0	29

Table 2: Sequential exclusion counts, charged only to the first successful criterion. Costs generally rise toward the bottom.

Criterion	$m = 1$	$m = 2$	$m = 3$	$m = 4$
Near-group theorem	2	1	1	1
Global dimension 15	1	0	0	0
Zero spectrum	8	30	51	63
Closed rank-four/five subring	0	25	125	235
Formal-codegree d-number	15	179	788	2314
Subcategory dimension quotient	0	5	9	8
Oriented dimension order	4	51	61	130

Table 3: Unitary model constructions identified by exact full-tensor based-ring isomorphism. Each row denotes a ring, not a count of categories.

Census ID	Explicit model
`R7_1_1`	$\mathrm{Rep}(D_{16})$
`R7_1_6`	$\mathrm{SU}(2)_6$
`R7_1_9`	$\mathrm{PSU}(2)_{12}$
`R7_1_12`	$\mathrm{SO}(7)_2$
`R7_1_16`	$\mathrm{Rep}(D_{22})$
`R7_1_17`	$\mathrm{PSU}(2)_{13}$
`R7_1_24`	$\mathrm{Zest}(\mathrm{SU}(2)_6)$
`R7_1_25`	$\mathrm{Rep}(SD_{16})$
`R7_1_39`	$\mathrm{TY}(C_6)$
`R7_1_43`	$\mathrm{Vec}(C_7)$
`R7_2_24`	$\mathrm{Rep}(S_5)$
`R7_2_233`	$\mathrm{Rep}(H_{24})$
`R7_2_234`	$\mathcal{C}(A_5, 1, A_4, 1)$
`R7_2_297`	$\mathrm{Rep}(SL_2(\mathbb{F}_3))$
`R7_2_319`	$\mathrm{Rep}(C_{13} \rtimes C_3)$
`R7_3_1059`	$\mathrm{Rep}(C_{11} \rtimes C_5)$

The first undecided multiplicity is one: the three remaining identifiers are $R7_1_19$, $R7_1_28$, and $R7_1_36$. The exhaustive source-position convention is $R7_m_j$, where j is the position *among rows of exact multiplicity* m in the pinned census file. This convention makes the positive-model permutations and negative witnesses unambiguous.

2 Necessary conditions and their order

The checks in Table 2 run from top to bottom on surviving tensors, and a tensor is charged to its first successful condition. Near-group and small-dimension checks read a few coefficients; the zero-spectrum and closed-subring passes are finite tensor scans; exact Casimir factoring and subcategory-dimension arithmetic follow; the complete oriented search is last. This order records precisely when a criterion first contributes rather than hiding it in an unordered union

of tests. Multiplicity one is already undecided, so these stages do *not* establish a multiplicity threshold beyond which no new criteria would be needed. The oriented criterion first contributes at multiplicity one, so it cannot be deferred to a higher multiplicity without losing a proof.

If G is the group of invertible basis elements and the only other object X satisfies $X^2 = \sum_{g\in G} g + \ell X$, then for $|G| = 6$: at $\ell = 0$ the Tambara–Yamagami construction forces G abelian [14]; at $0 < \ell < 5$ in the commutative case, Schopieray's near-group obstruction rules out categorification [13, Proposition 3.2]. These statements apply to a categorification without a pivotal assumption. The certificate records the six group elements, the remaining object and ℓ, and rechecks every fusion-rule hypothesis.

The global-dimension-fifteen branch has six dimension-one simples and one dimension-three simple: direct substitution into every product establishes that $(1, \dots, 1, 3)$ is its strictly positive dimension character. Hence its Frobenius–Perron dimension is $6 + 3^2 = 15$. The *pq* classification [6, Theorems 3.1 and 5.1] makes a complex fusion category of dimension $3 \cdot 5$ pointed in this case, contradicting the dimension-three simple.

The zero-spectrum obstruction is the exact necessary criterion of Liu–Palcoux–Ren [9]; each certificate stores its nine basis indices and the verifier rechecks all required zero/nonzero fusion coefficients. A proper fusion-closed subset of basis elements would generate a fusion subcategory. If its induced rank-four or rank-five ring is excluded by the complete lower-rank classifications [2, 3], the ambient ring is excluded. The imported model registries contain all fifteen rank-four and all seventeen rank-five realized based-ring types, checked up to multiplicities 128 and 20, respectively; all encountered subrings fall within these ranges. The complete rank-six multiplicity-two classification [4] supplies context and methodology, without being used as an unstated rank-seven classification theorem.

Proposition 2 (The formal-codegree test at rank seven)**.** *Suppose that R is a rank-seven fusion ring, possibly noncommutative. Let $C = \sum_{i=0}^{6} N_i N_i^{\mathsf{T}}$ be its regular Casimir matrix. If an irreducible factor of the characteristic polynomial of C is not the minimal polynomial of a d-number, then R cannot be the Grothendieck ring of a complex fusion category.*

Proof. In the commutative case, the eigenvalues of C are the formal codegrees of one-dimensional characters and must be d-numbers by Ostrik [11]. In the noncommutative case, semisimplicity, $\dim_{\mathbb{C}}(R \otimes \mathbb{C}) = 7$, and the positive one-dimensional character imply the unique possible decomposition $R \otimes \mathbb{C} \simeq \mathbb{C}^3 \oplus M_2(\mathbb{C})$. There is only one two-dimensional irreducible module; it is fixed by every field automorphism, so its formal codegree is a rational algebraic integer and hence an integer. The Casimir acts on this matrix block by twice that integer. Every remaining nonrational Casimir eigenvalue comes from a one-dimensional character and is a formal codegree, so Ostrik's condition applies. The implementation factors the exact integral characteristic polynomial and tests the coefficient-divisibility characterization of d-numbers. The replay checker recomputes and factors that polynomial. □

If S is a proper commutative fusion subring of a hypothetical categorification, the quotient $\mathrm{FPdim}(R)/\mathrm{FPdim}(S)$ must be an algebraic integer [7]. The subcategory-divisibility stage computes this quotient in the exact Frobenius–Perron number field, records its characteristic polynomial, and certifies a nonintegral coefficient. Its certificate is rederived from the complete tensor.

3 Complete oriented tests at rank seven

The last row of Table 2 uses the general, *oriented* signed-sphericalization construction established in [4]. For an arbitrary hypothetical category $\mathcal{C}$, its canonical sphericalization has two lifts of

each simple object, interchanged by a sign object ε. Taking the quotient by $(\varepsilon + 1)$ produces integer coefficients T_{ij}^k with $|T_{ij}^k| \leq N_{ij}^k$ and $T_{ij}^k \equiv N_{ij}^k \pmod 2$. Unit, duality, cyclic Frobenius reciprocity, and ordered associativity further constrain the quotient. The self-dual positive-indicator normalization imposes the established square-channel congruences modulo four. These conditions are necessary and cover every categorification, whether pivotal or not; a signed solution is not itself a categorification.

The rank-six proof of the general construction and the dimension-order obstruction [4] are rank-independent. Here their enumeration engine is extended to seven basis elements. It enumerates the independent cyclic/dual-reflection coefficient orbits, prunes only exact associativity equations with checked integer bounds, and constructs *every* signed quotient. For each quotient the checker verifies either its adjoint parity, its gauge equivalence with the original ring, or the failure of every admissible spherical character by codegree and dimension-order tests. In particular the dimension-order test requires the central-induction dimension δ/f_χ to lie in the $\mathbb{Z}$-span of the signed simple dimensions, as required by [12, 4].

Proposition 3 (Verified exhaustive branches)**.** *The oriented-dimension-order criterion excludes four, fifty-one, sixty-one, and one hundred thirty previously surviving rank-seven rings of exact multiplicities one through four, respectively. For* R7_1_5*,* R7_1_23*,* R7_1_33*, and* R7_1_38*, the complete signed enumerations have respectively* $2, 4, 4, 8$ *associative quotient tables. Every table in each of these 246 certified exhaustive searches is eliminated by necessary conditions for arbitrary complex fusion categories.*

Proof. The machine-readable proof tree for each of the 246 rings gives its full dimension-character factorization, every obstruction, the ordered signed-orbit domain, and one checked branch for every enumerated table. The independent verifier rebuilds the domains from the source tensor, repeats the bounded exhaustive enumeration, checks the *entire* list of surviving tables for equality to the recorded list, and recomputes each branch witness. The search used to discover these finite trees is logically separate from their exhaustive replay; timed-out or unfinished searches generate no negative decision. The enumeration returns the four displayed multiplicity-one branch counts and no uncovered branch for any of the 246 certificates. The rank-independent necessity of the search is the sphericalization theorem cited above. □

For the three surviving multiplicity-one rows the current general filters provide no exclusion. Publicly proposed numerical categorical data [16, 17] provide useful leads; no unverified numerical solution is promoted here to a constructive model. This same standard applies to the larger residual sets at multiplicities two, three, and four.

4 Exact constructions and replay

The positive model catalogue constructs complete fusion tensors from the cyclic group, ordinary finite-group character products, the Tambara–Yamagami rule, $SU(2)_k$ truncation, its even parts, odd metaplectic rules, and the categorical $\mathbb{Z}/2$ zest of $SU(2)_6$ [14, 8, 10, 15, 16]. Character products for $\mathrm{Rep}(S_5)$, $\mathrm{Rep}(SL_2(\mathbb{F}_3))$, $\mathrm{Rep}(C_{13} \rtimes C_3)$ and $\mathrm{Rep}(C_{11} \rtimes C_5)$ are computed with exact characters in the bundled software. Two further complete model tensors come from the published noncommutative integral appendix of [5, Theorem 1.6 and Section 6.2]: the representation category of its dimension-24 Kac algebra and the group-theoretical category $\mathcal{C}(A_5, 1, A_4, 1)$. The published full tensors and their byte digest are pinned in the package. For each constructed or published model, the program validates the fusion axioms, then checks all 6! unit-preserving basis permutations

against every compatible census tensor. A positive record carries its explicit permutation. Different categorical structures may share a based ring; the count is of rings, not categories.

The complete package contains the pinned gzip source, analysis code, negative proof trees, model tensors, the stage-count summary and a second entry point, `verify.py`. Run `python analyze.py -max-m 4 -dnumber` and then `python verify.py`. The verifier rechecks the source digest and every row, every full-tensor model isomorphism, each stage total and each exclusion certificate, including the exhaustive signed lists. Running `python -O verify.py` gives the same checks without relying on Python's optional `assert` statements.

AI disclosure. GPT-6 Astra assisted with research, symbolic exploration, code development, and drafting.

References

[1] S. Palcoux, *Fusion Ring Census*, https://github.com/sebastienpalcoux/fusion-ring-census, commit `b86eb6dc6eeeafe5fc5f6906d7b074e0322e9726`.

[2] *Rank-four Grothendieck rings through multiplicity 128*, manuscript and exact-certificate supplement (2026).

[3] *Rank-five Grothendieck rings through multiplicity twenty*, manuscript and exact-certificate supplement (2026).

[4] *Rank-six fusion rings: categorification criteria and models through multiplicity eight*, manuscript and exact-certificate supplement (2026).

[5] M. A. Alekseyev, W. Bruns, J. Dong and S. Palcoux, *Classifying integral Grothendieck rings up to rank 5 and beyond*, accepted in *Letters in Mathematical Physics*; arXiv:2507.07023.

[6] P. Etingof, S. Gelaki and V. Ostrik, *Classification of fusion categories of dimension pq*, *Int. Math. Res. Not.* **2004**, 3041–3056; arXiv:math/0304194.

[7] P. Etingof, D. Nikshych and V. Ostrik, *On fusion categories*, *Ann. of Math.* **162** (2005), 581–642; arXiv:math/0203060.

[8] A. Kirillov Jr. and V. Ostrik, *On a q-analogue of the McKay correspondence and the ADE classification of $\widehat{\mathfrak{sl}}_2$ conformal field theories*, *Adv. Math.* **171** (2002), 183–227; arXiv:math/0101219.

[9] Z. Liu, S. Palcoux and Y. Ren, *Classification of Grothendieck rings of complex fusion categories of multiplicity one up to rank six*, *Lett. Math. Phys.* **112** (2022), article 54; arXiv:2010.10264.

[10] P. Bruillard, C. Galindo, S.-H. Ng, J. Y. Plavnik, E. C. Rowell and Z. Wang, *On the classification of weakly integral modular categories*, *J. Pure Appl. Algebra* **220** (2016), 2364–2388; arXiv:1411.2313.

[11] V. Ostrik, *On formal codegrees of fusion categories*, *Math. Res. Lett.* **16** (2009), 895–901; arXiv:0810.3242.

[12] V. Ostrik, *Pivotal fusion categories of rank 3*, *Mosc. Math. J.* **15** (2015), 373–396; arXiv:1309.4822.

[13] A. Schopieray, *Categorification of integral group rings extended by one dimension*, *J. Lond. Math. Soc.* **108** (2023); DOI:10.1112/jlms.12789; arXiv:2208.07319.

[14] D. Tambara and S. Yamagami, *Tensor categories with fusion rules of self-duality for finite abelian groups*, *J. Algebra* **209** (1998), 692–707.

[15] C. Delaney, C. Galindo, J. Y. Plavnik, E. C. Rowell and Q. Zhang, *Braided zesting and its applications*, *Comm. Math. Phys.* **386** (2021), 1–55; arXiv:2005.05544.

[16] G. Vercleyen, *On low-rank multiplicity-free fusion categories*, thesis (2024), arXiv:2405.20075.

[17] G. Vercleyen, *Tables of practical invariants for distinguishing multiplicity-free fusion categories up to rank 7*, arXiv:2507.00652.